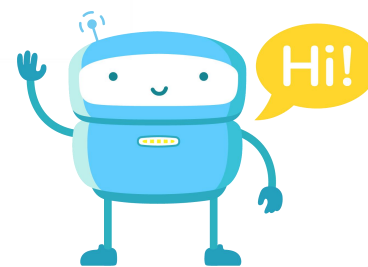


自然语言处理

第二章：机器学习与神经网络基础（二）

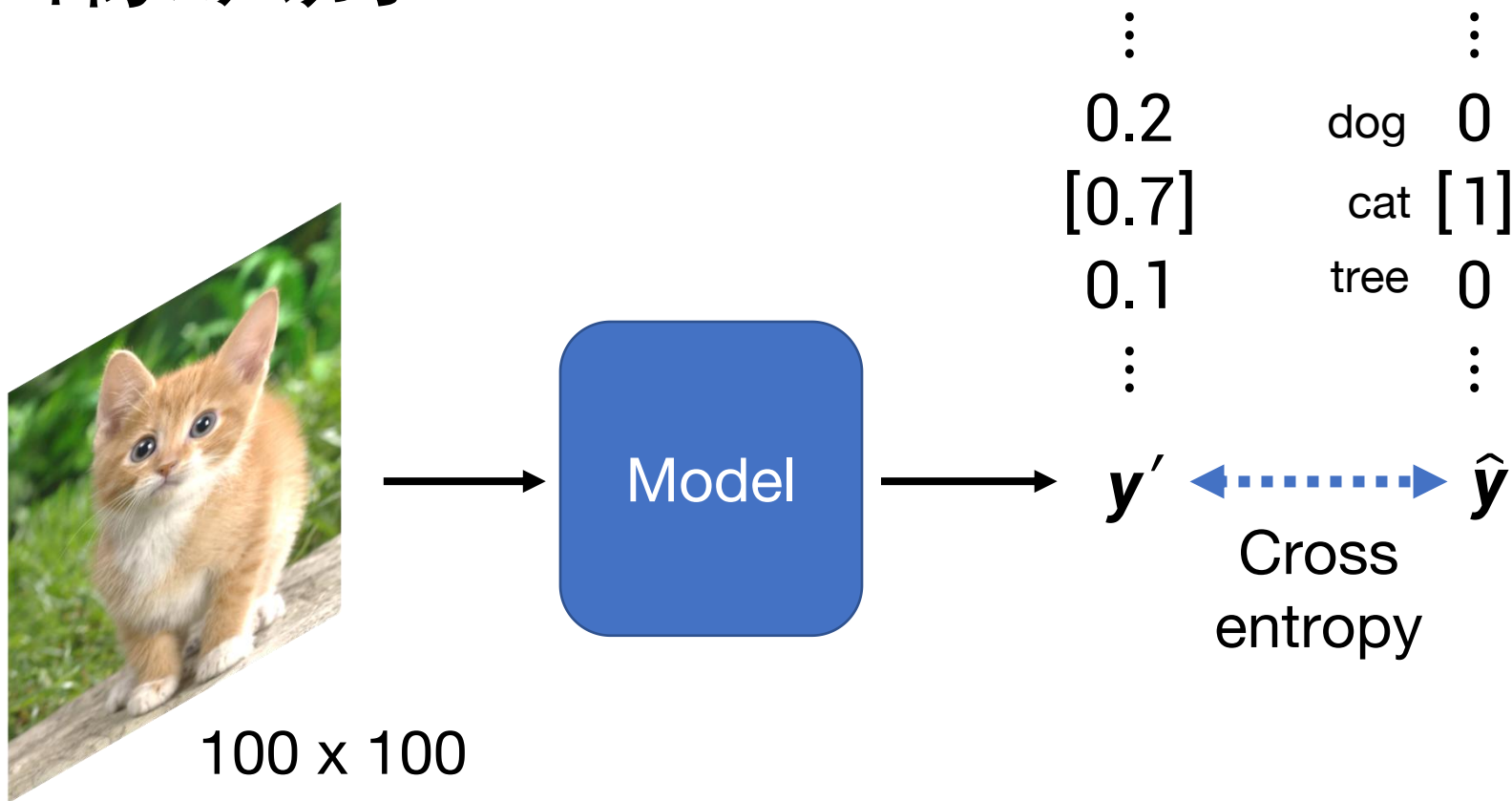


授课老师： 缪尧



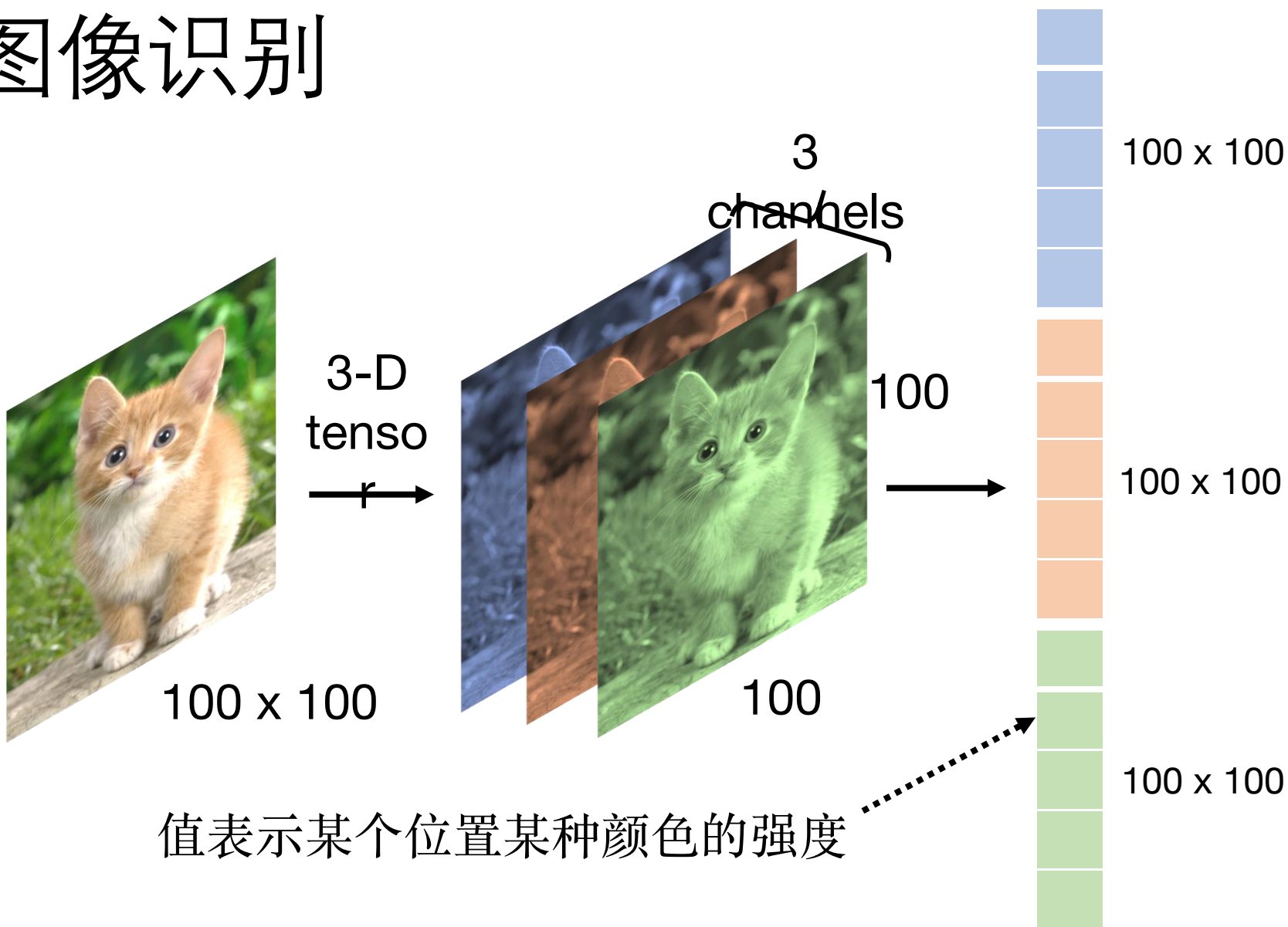
注意：本PPT仅供个人学习使用，任何组织和个人不得公开传播或用于任何商业盈利用途，否则一切后果由该组织或个人承担。

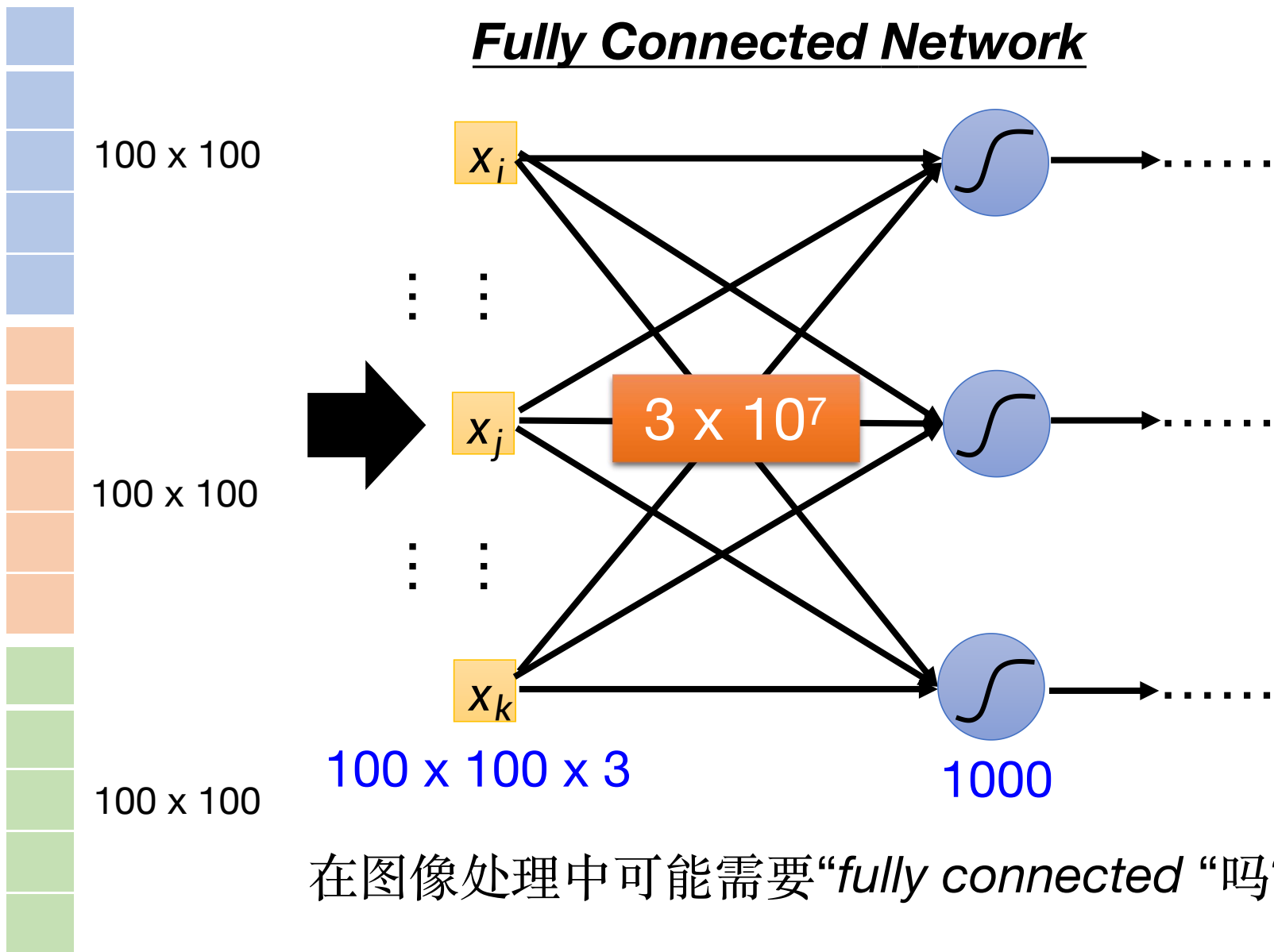
图像识别



(所有被用来分类的图片都具有同样的 size.)

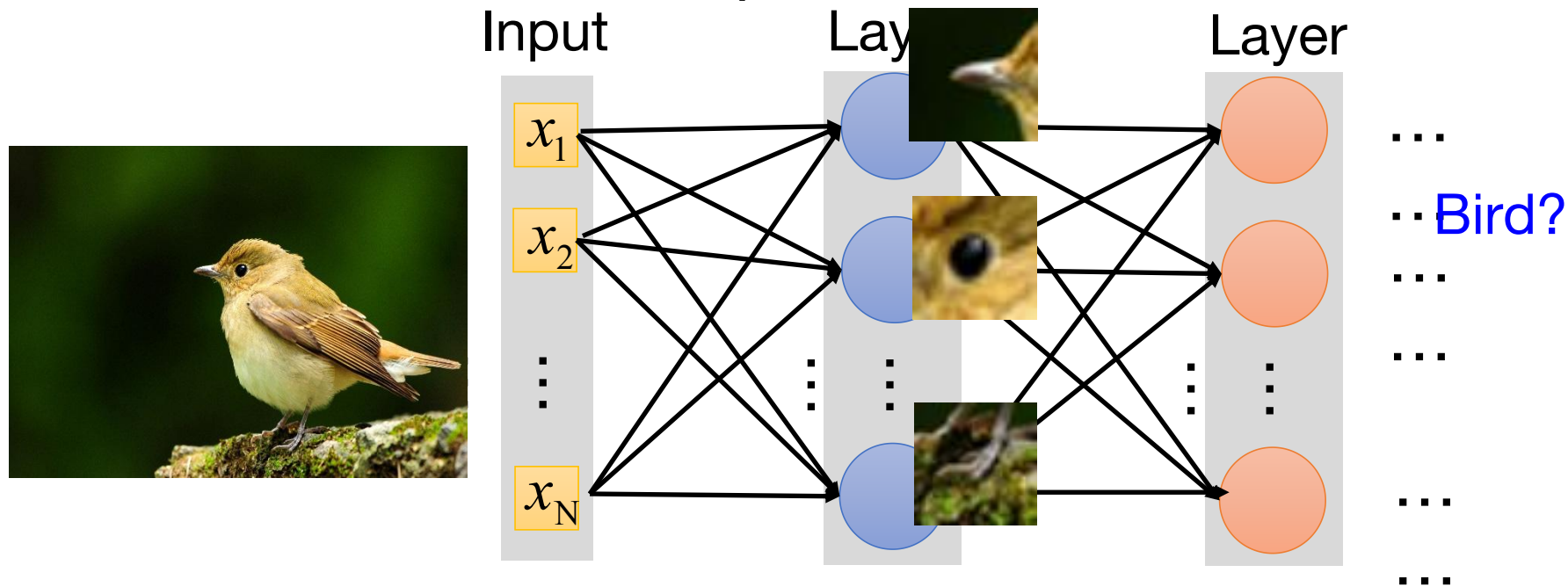
图像识别





在图像处理中可能需要“*fully connected*”吗？

举例 1

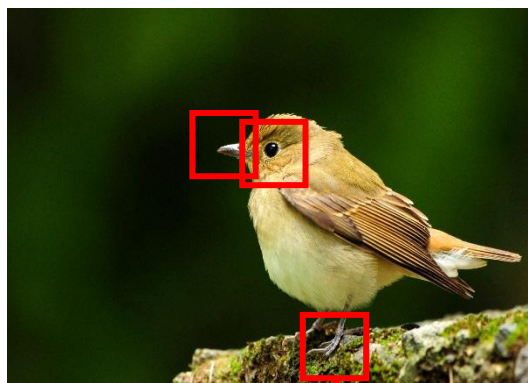


可能人类也是用同样的方式去识别图片中的鸟... 😊



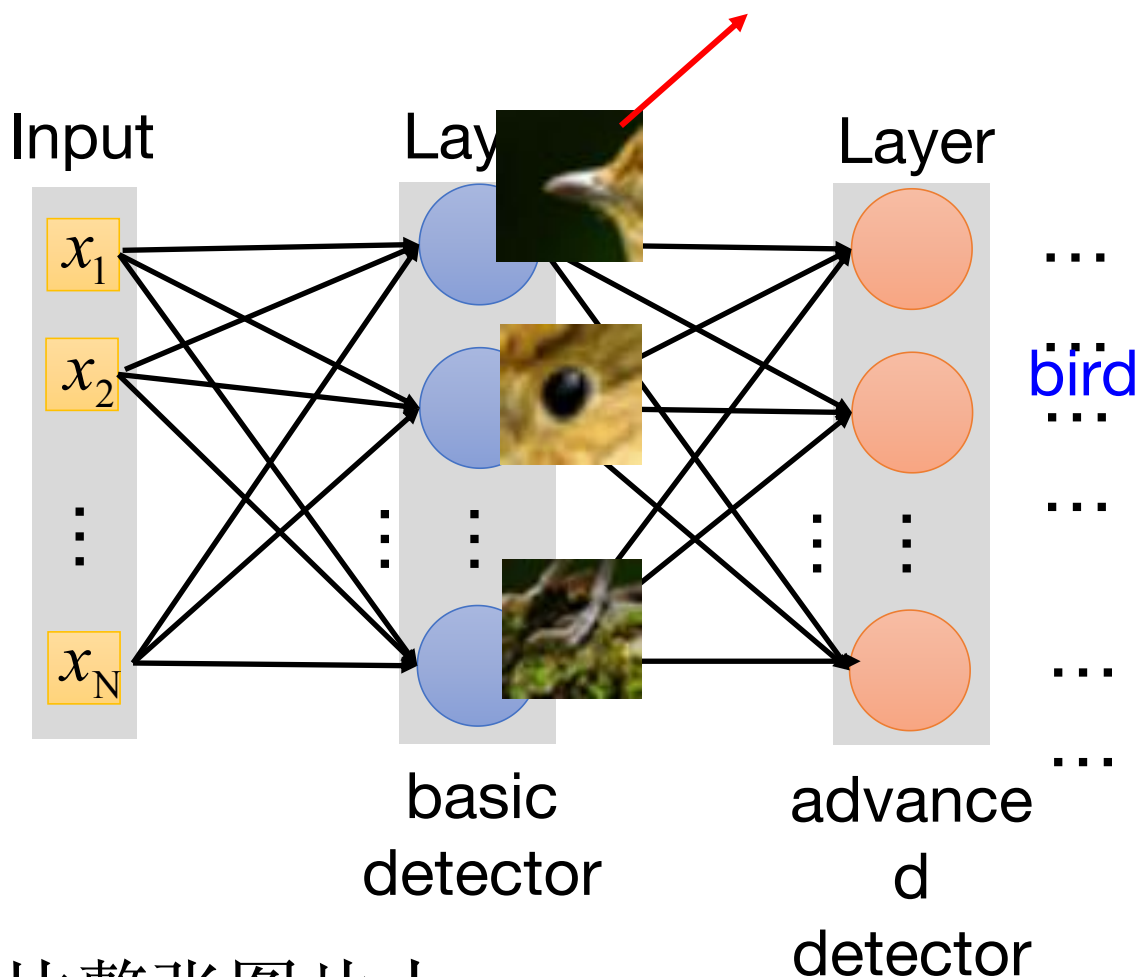
举例 1

需要看整张图片吗?

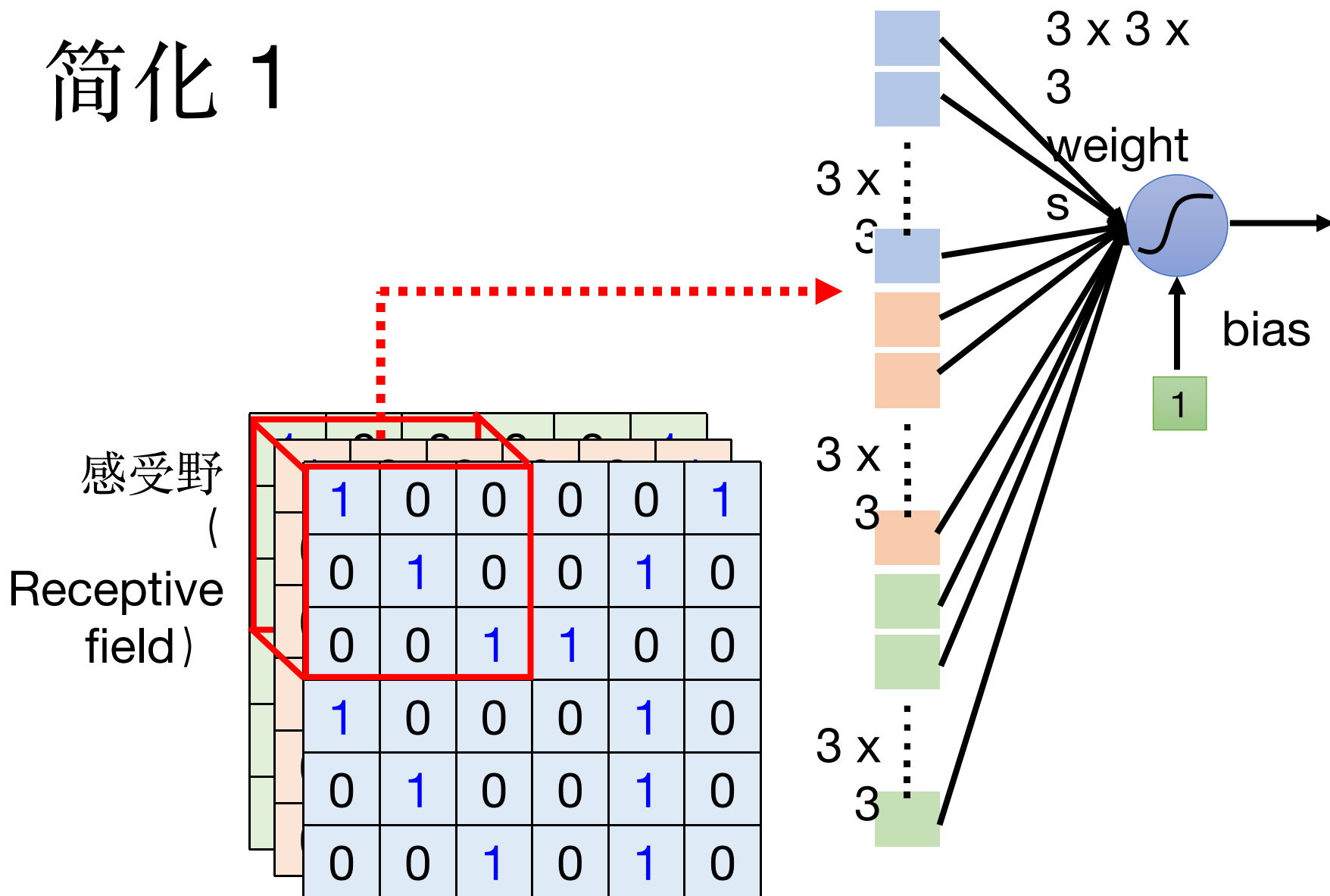


某些patterns比整张图片小.

每个neuro不需要看整张图片.

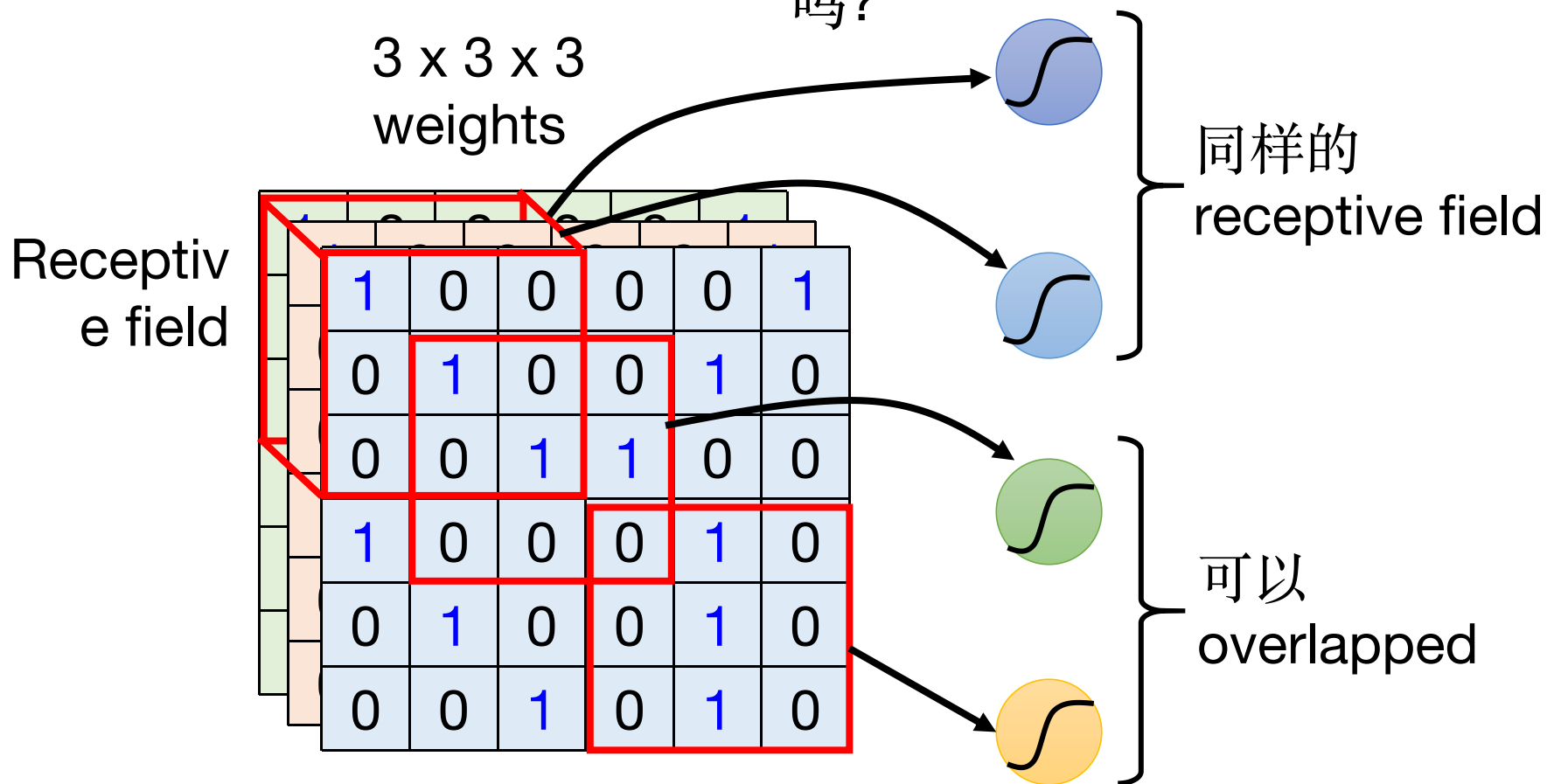


简化 1



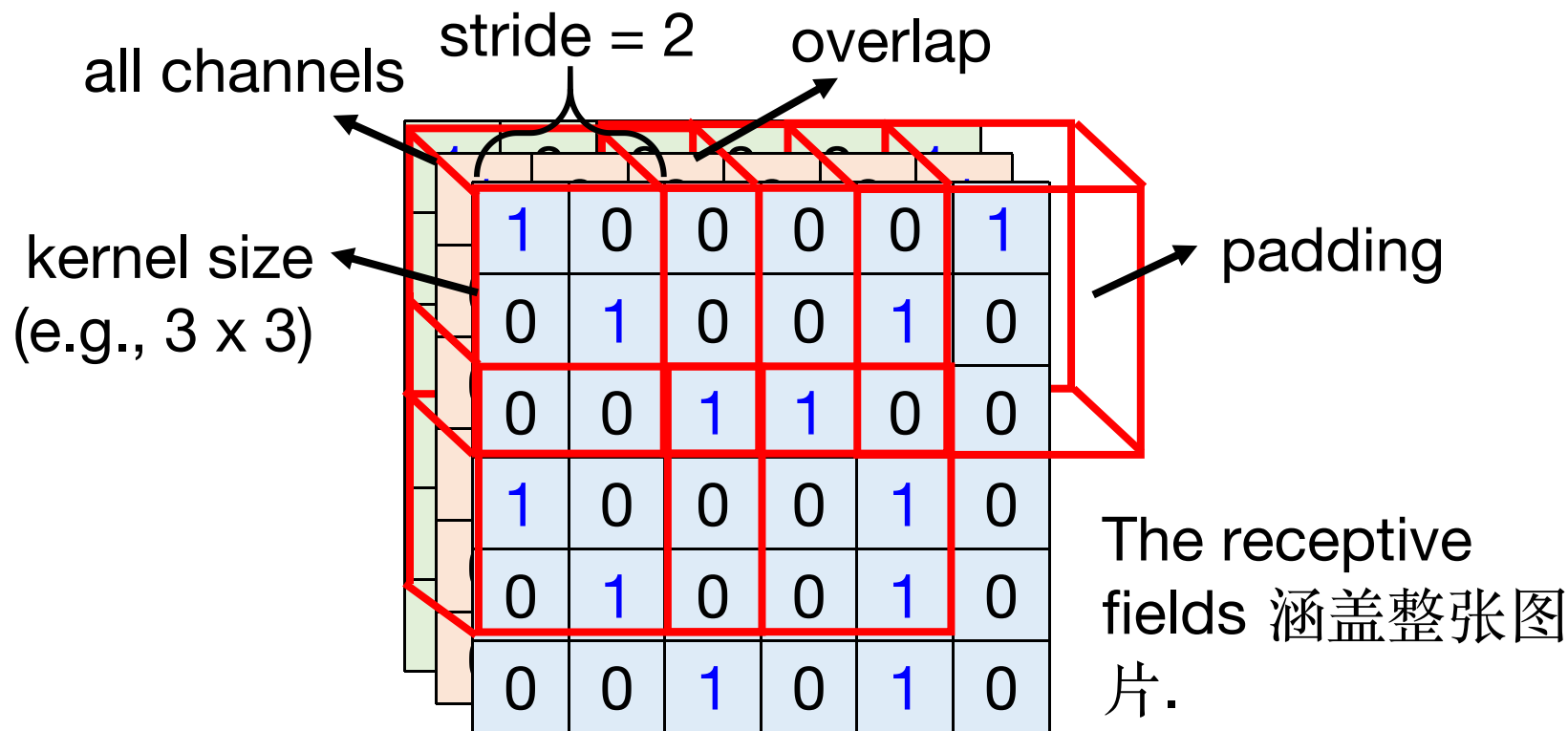
简化1

- 不同的neurons可以有不同尺寸的receptive field吗?
- 可以只涵盖一些channels吗?
- receptive field可以不是正方形吗?



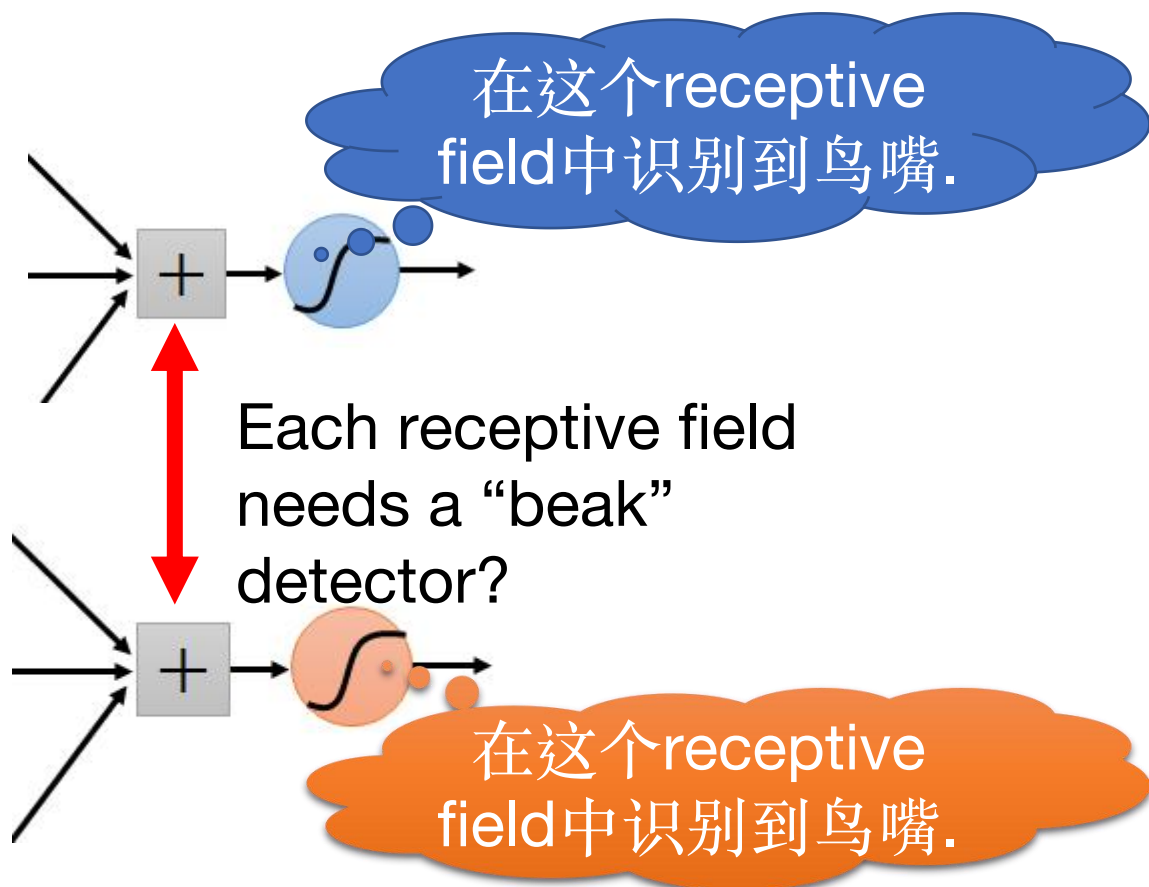
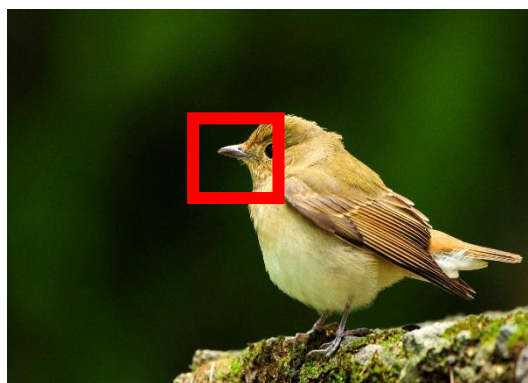
简化 1 – 典型的设置

每一个receptive field 有一组neurons (e.g., 64 neurons).

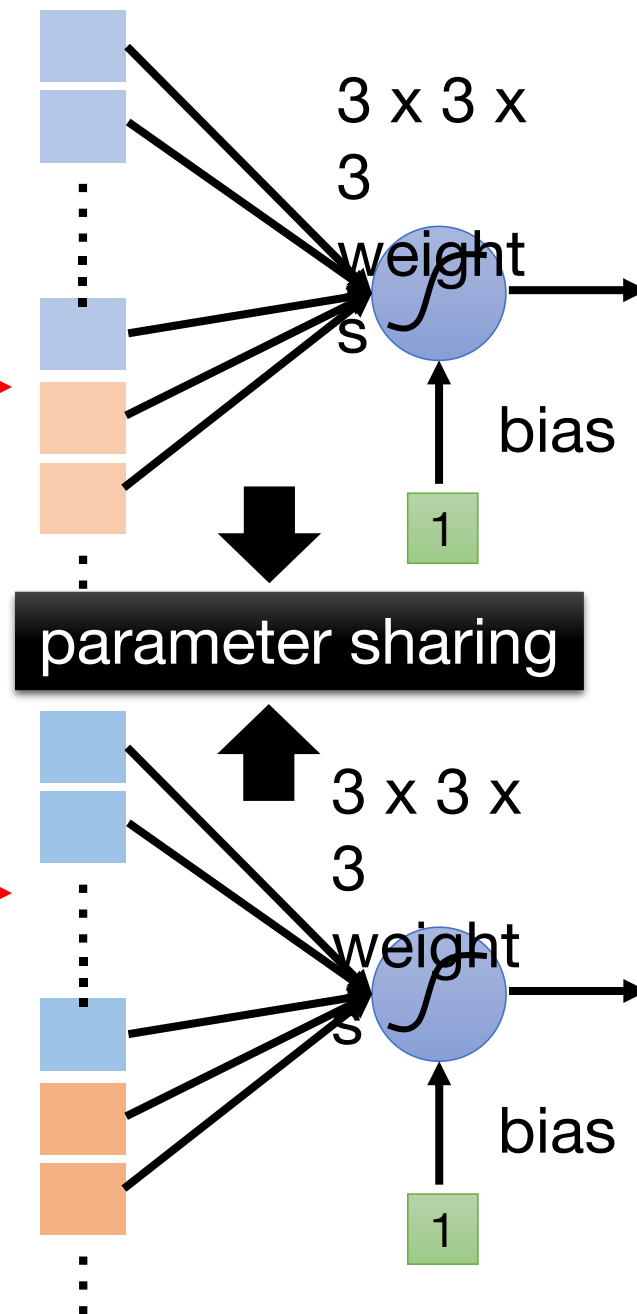
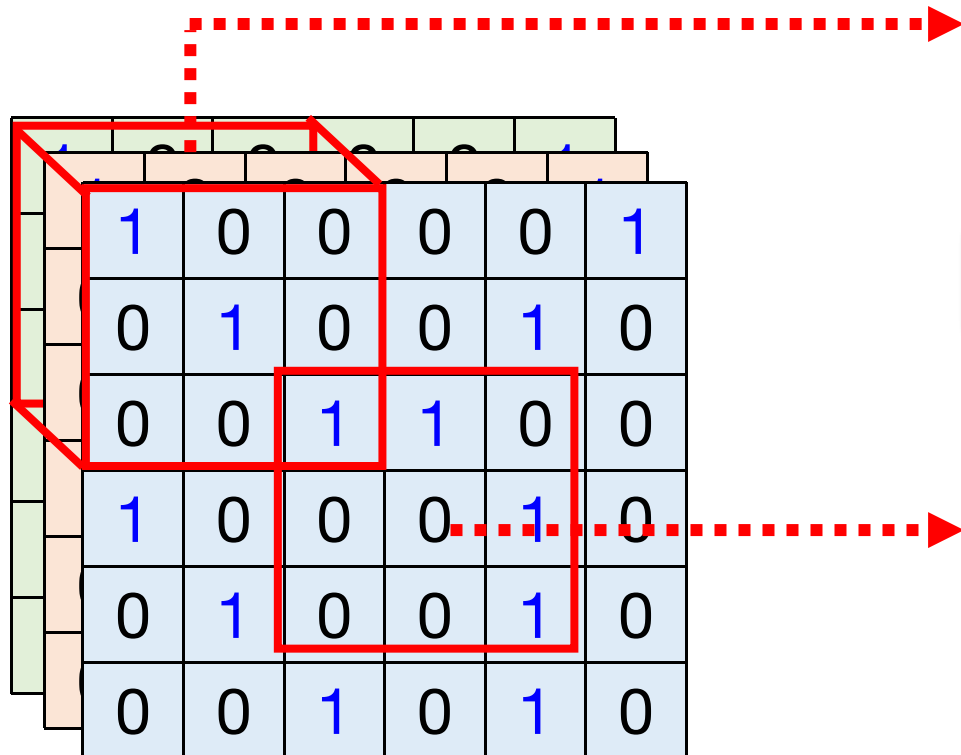


举例 2

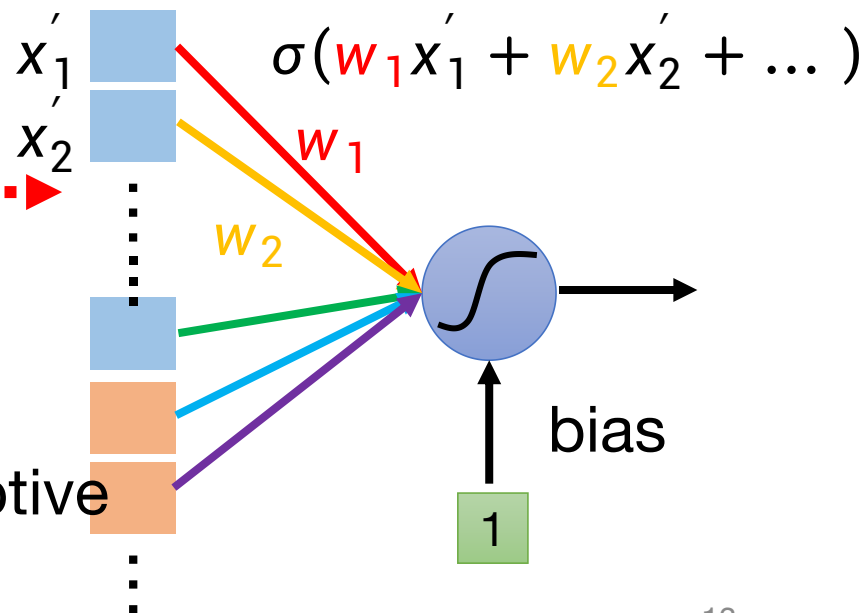
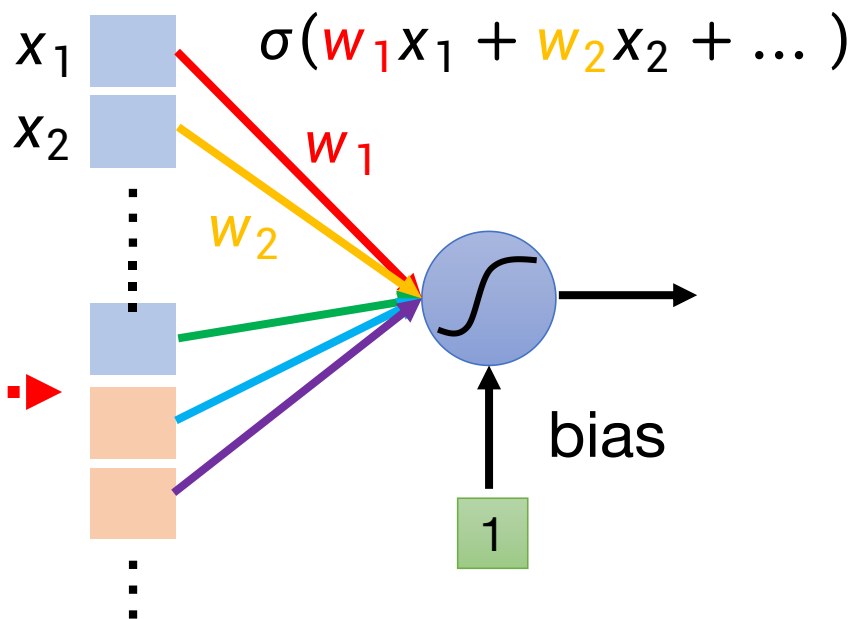
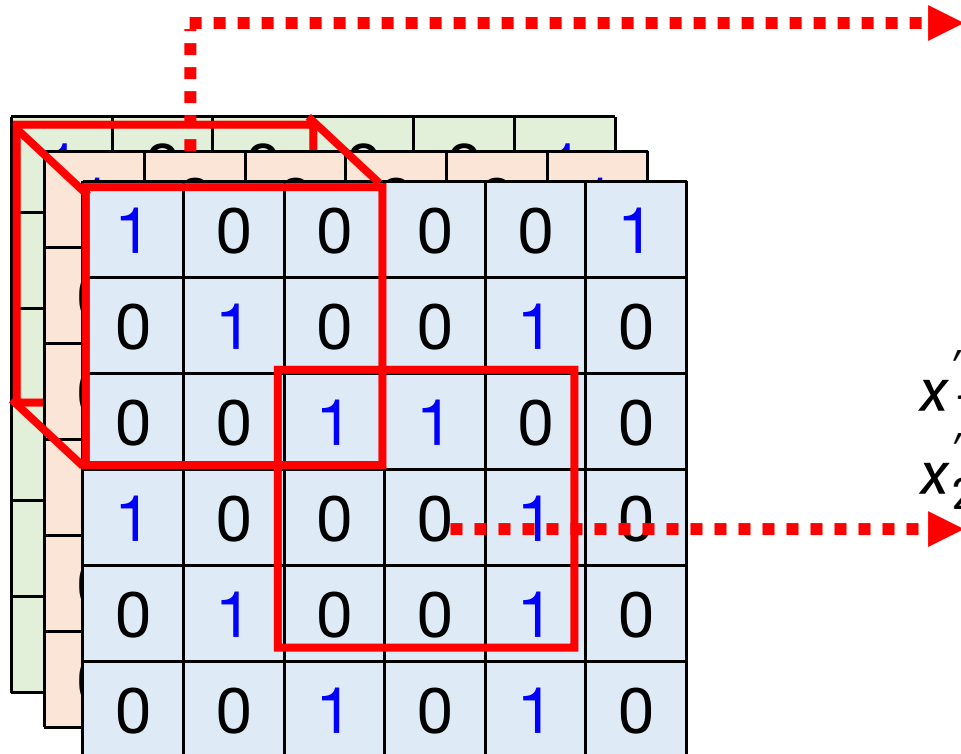
- 同样的patterns 出现在不同的区域.



简化 2



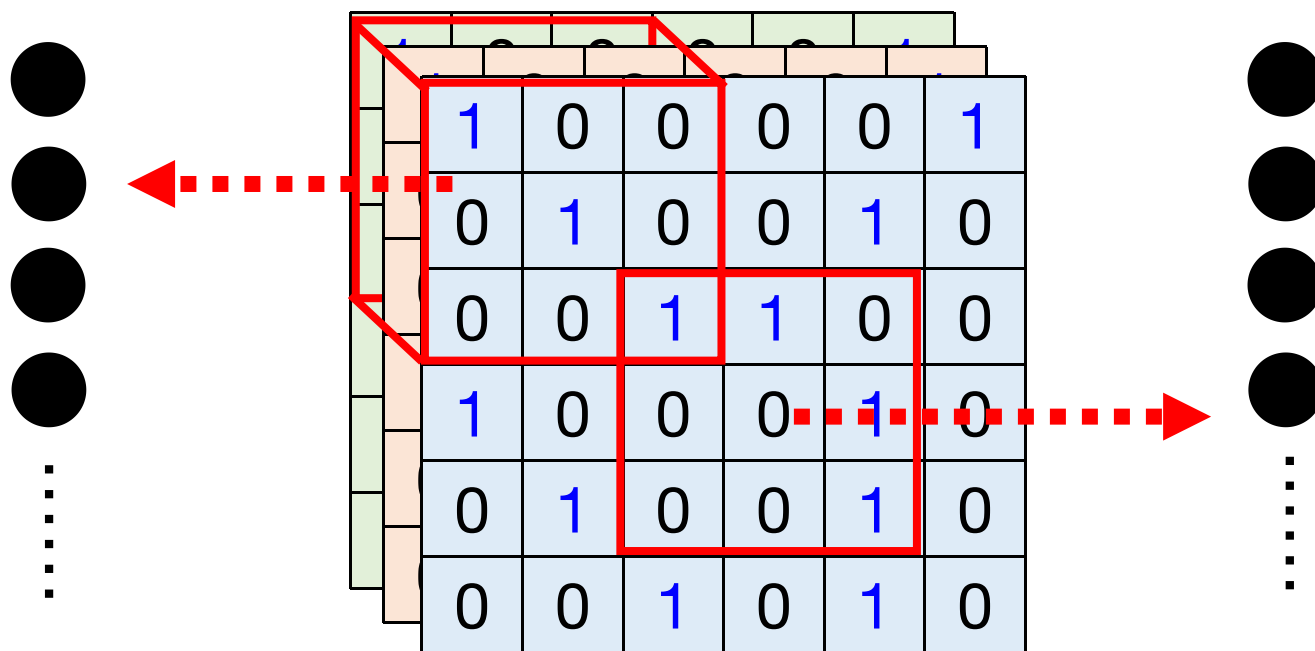
简化 2



Two neurons with the same receptive field would not share parameters.

简化 2 - 典型设置

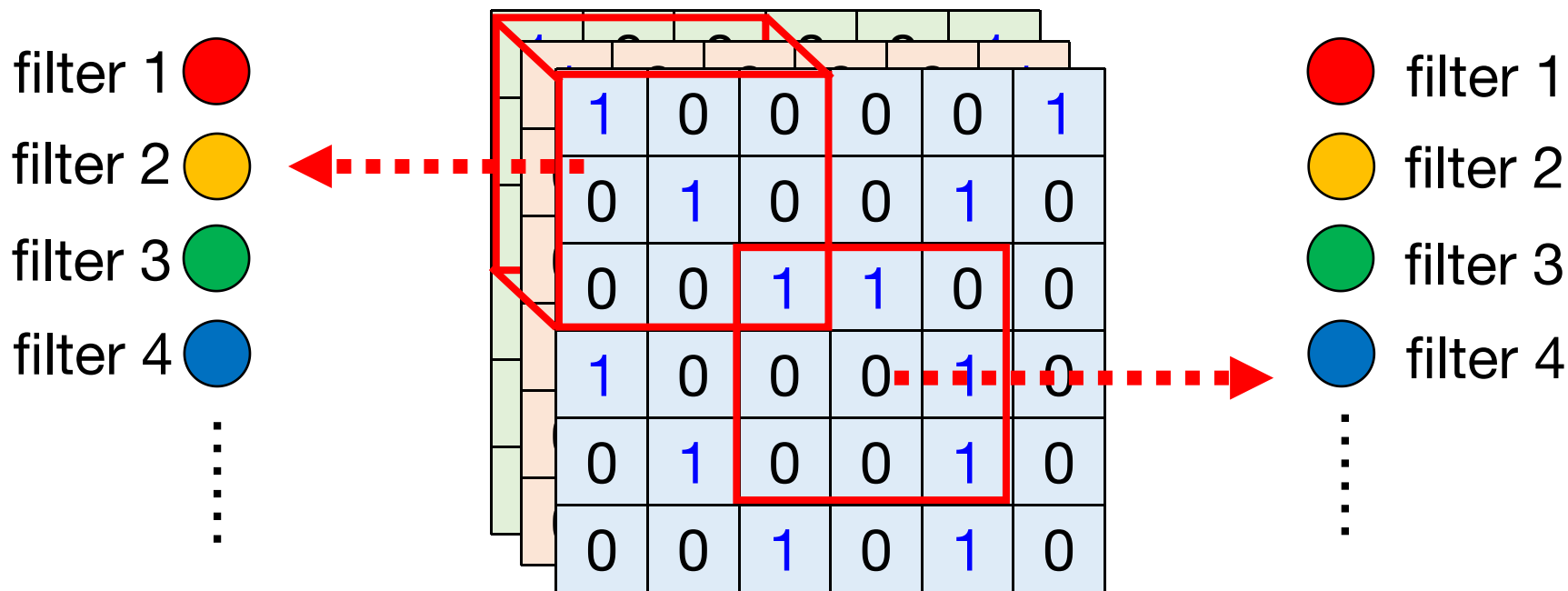
每一个receptive field 都有一组neurons (e.g., 64 neurons).



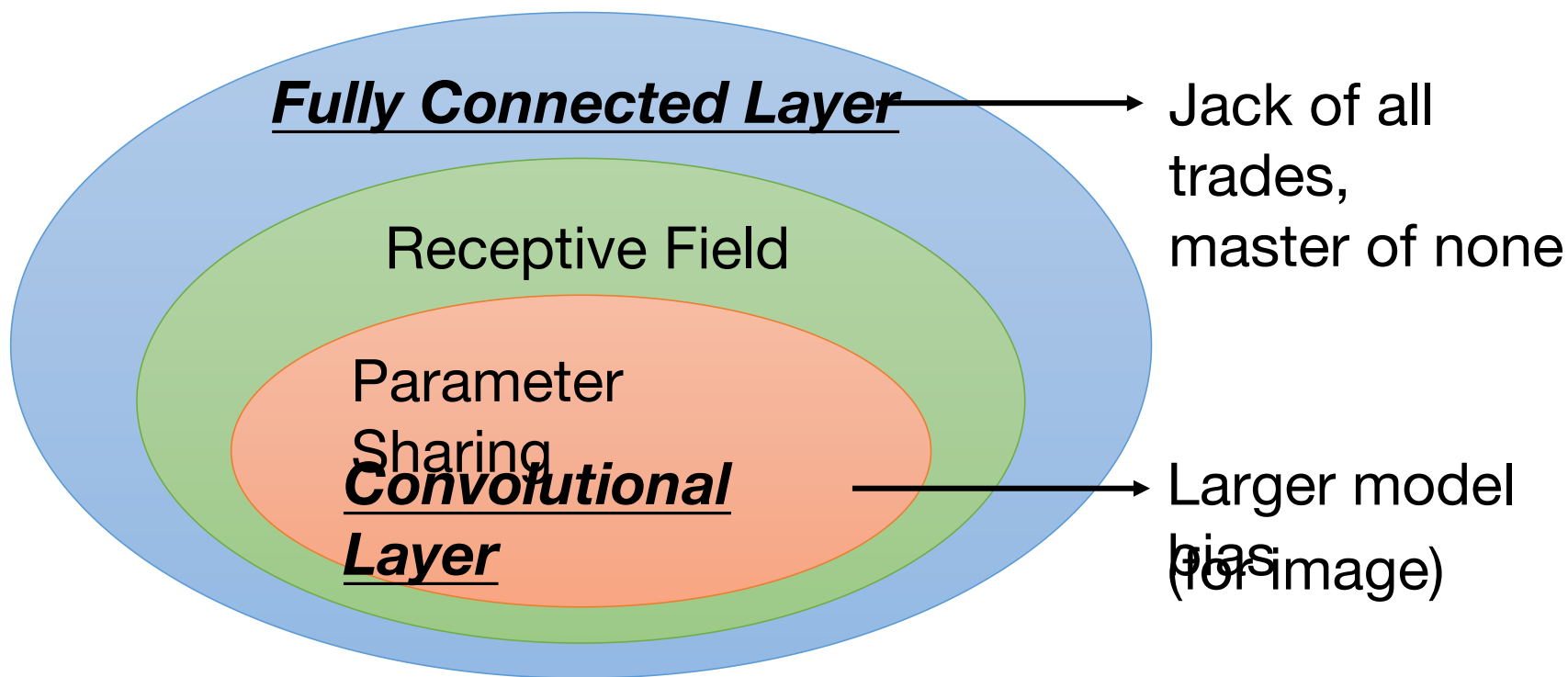
简化 2 - 典型设置

每个receptive field只有一组neurons (e.g., 64 neurons).

每个 receptive field has the neurons with the same set of parameters.



Convolutional Layer 的优势



- 一些patterns 比整张图片小.
- 相同的patterns出现在不同的区域.

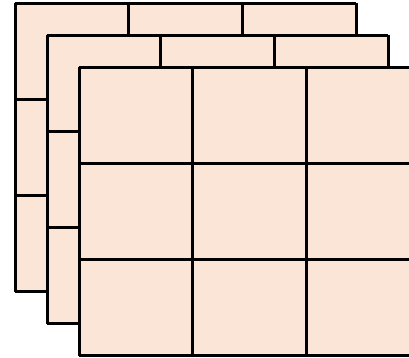
Another story based on *filter*

Convolutional Layer ☺

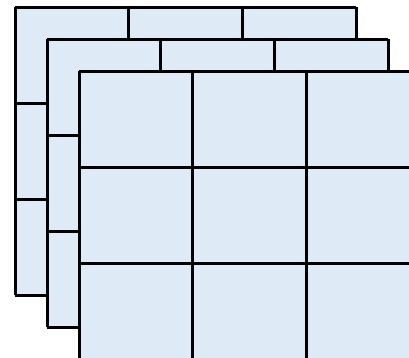


Convolution

⋮



Filter 1
3 x 3 x
channel



Filter 2
3 x 3 x
channel

⋮

channel = 3(colorful)

channel = 1(black and white)

Each filter detects a small pattern (3 x 3 x channel).

Convolutional Layer

Consider channel = 1
(black and white
image)

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

⋮ ⋮

(The values in the filters
are unknown
parameters.)

Convolutional Layer

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

Convolutional Layer

-1	1	-1
-1	1	-1
-1	1	-1

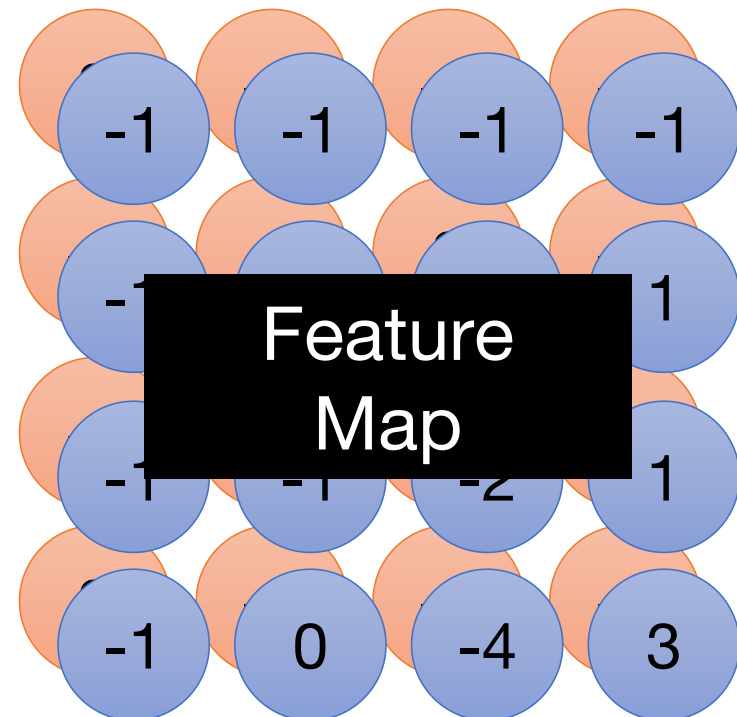
Filter 2

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Do the same process
for every filter



Convolutional Layer

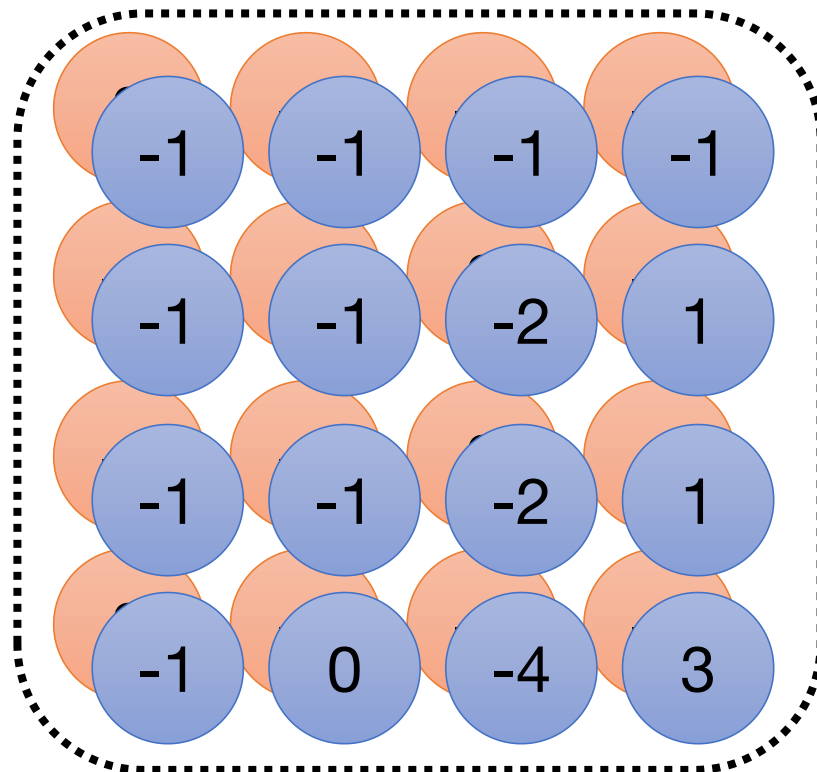


64
filters

Convolution
n

Convolution
n

⋮
⋮



“Image” with 64 channels

Multiple Convolutional Layers

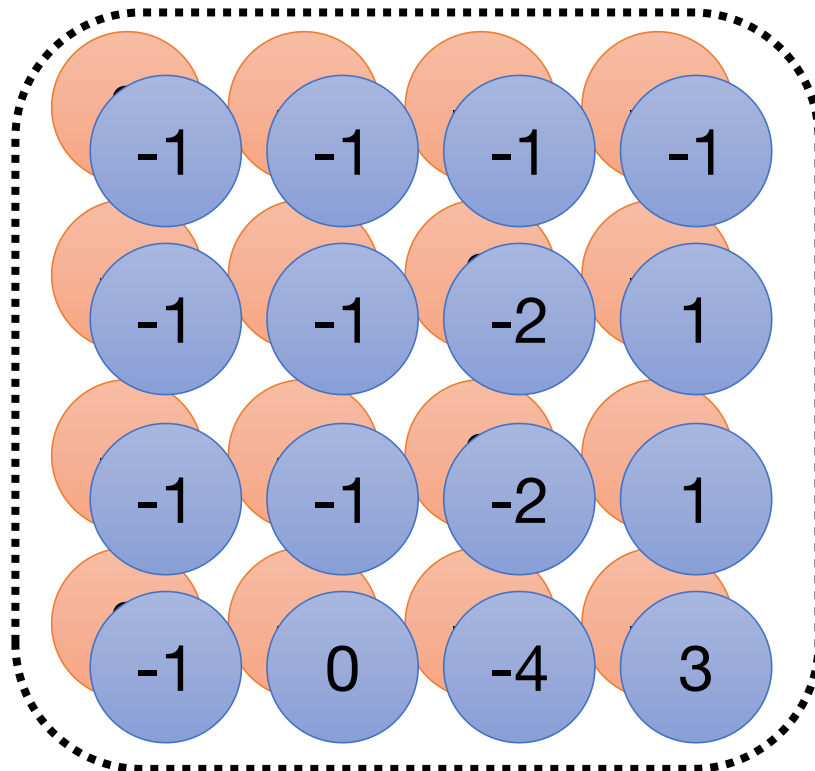


64
filters

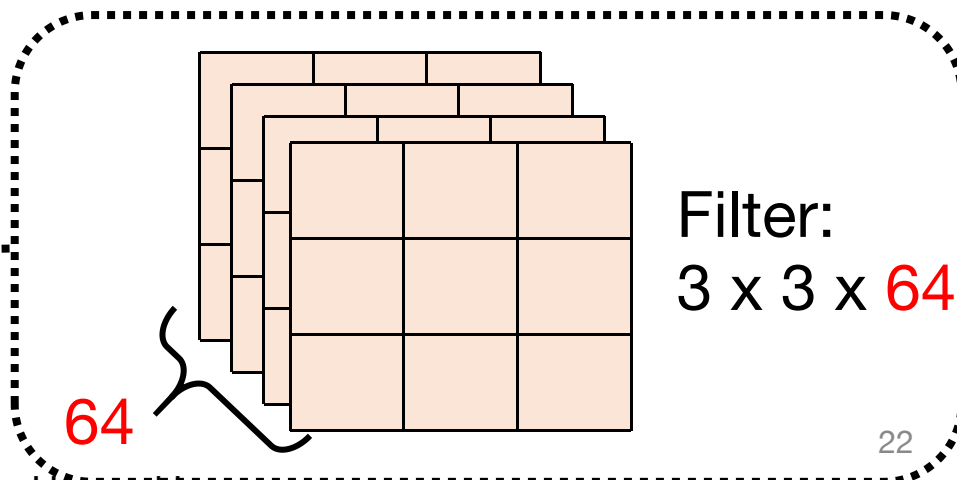
Convolutio
n

Convolutio
n

⋮
⋮



“Image” with 64 channels



Multiple Convolutional Layers



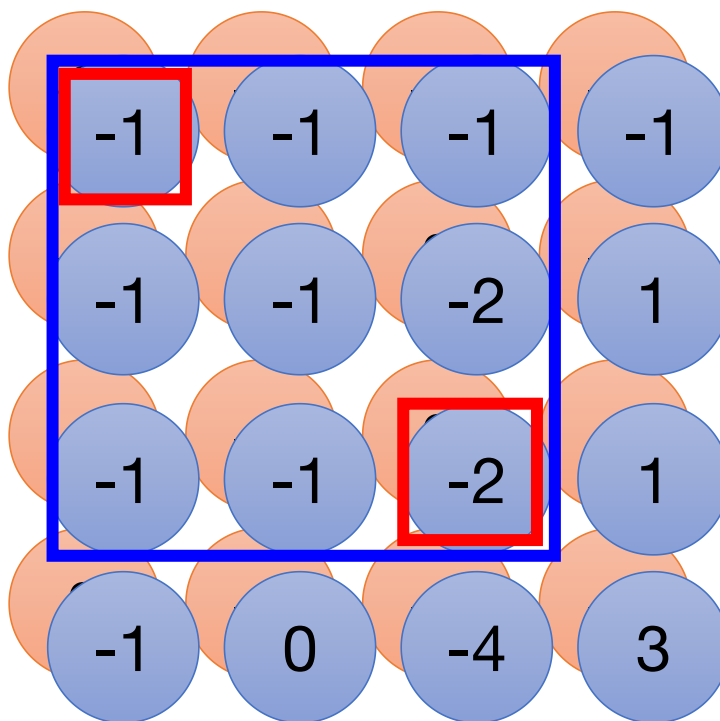
64
filters

Convolution

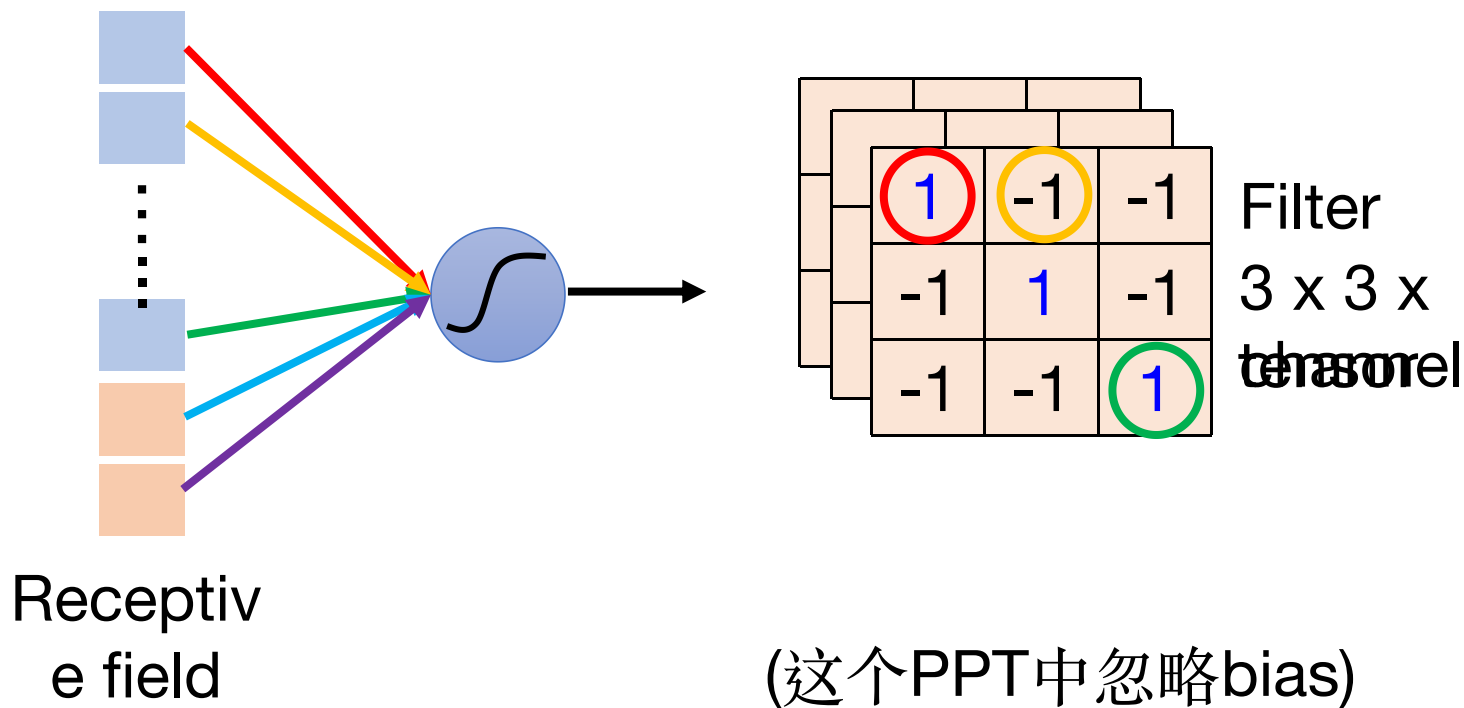
Convolution

⋮
⋮

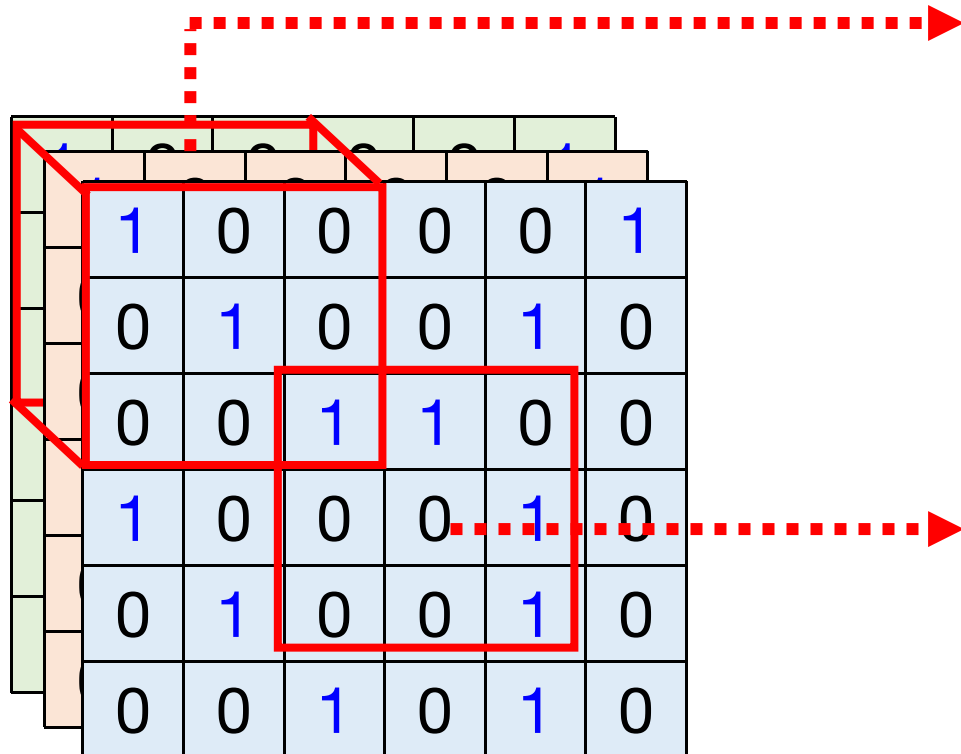
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0



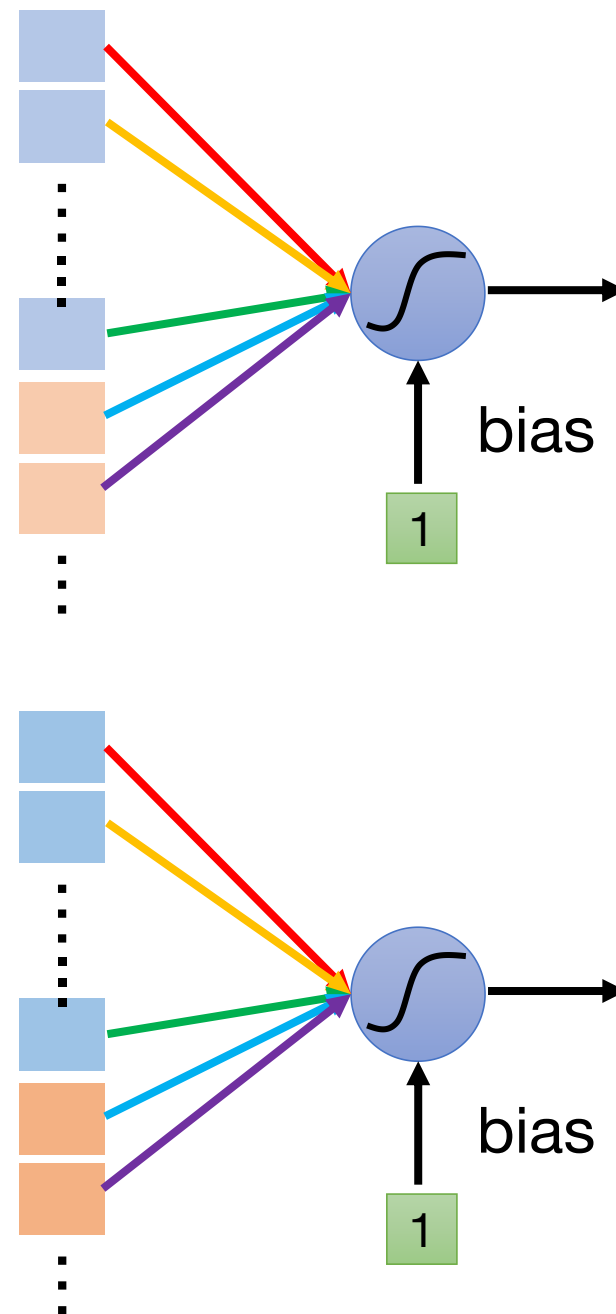
Comparison of Two Stories



不同receptive fields的neuro共享参数.



Each filter convolves over the input image.



示例

- Subsampling the pixels will not change the object

bird



subsampling

bird



Pooling – Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

Convolutional Layers + Pooling

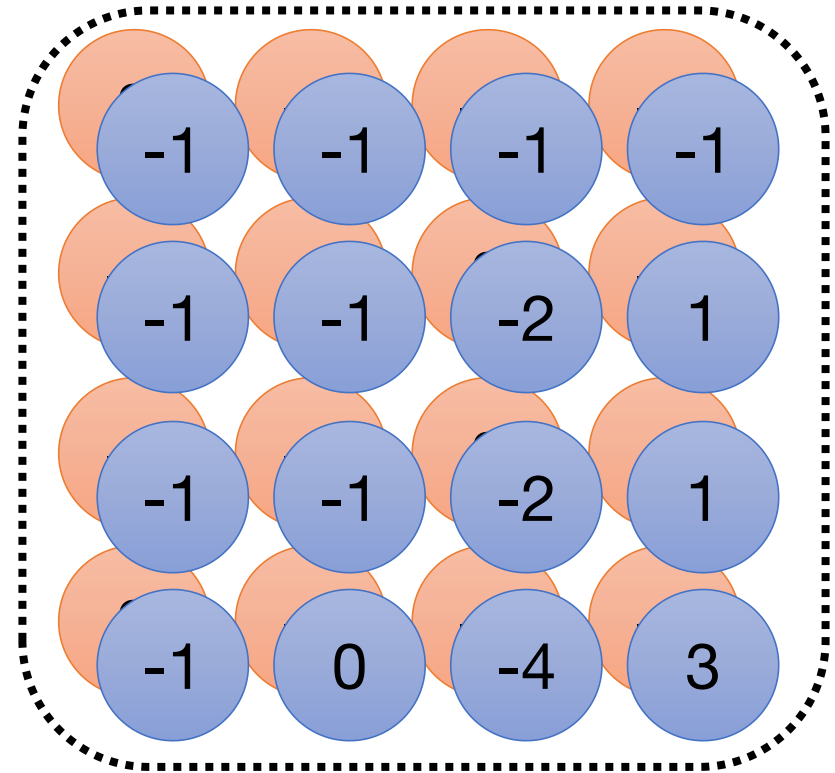


Convolution

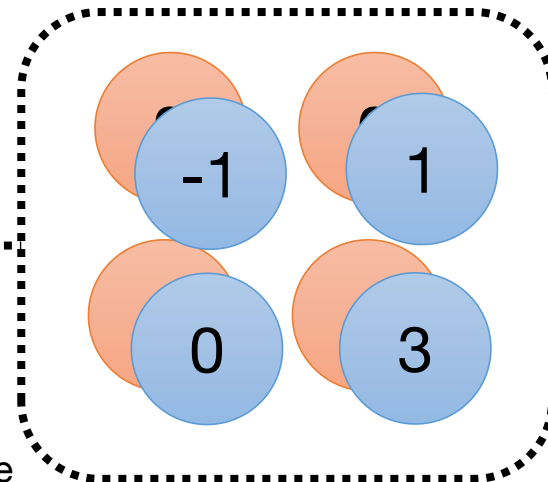
Pooling

Repeat

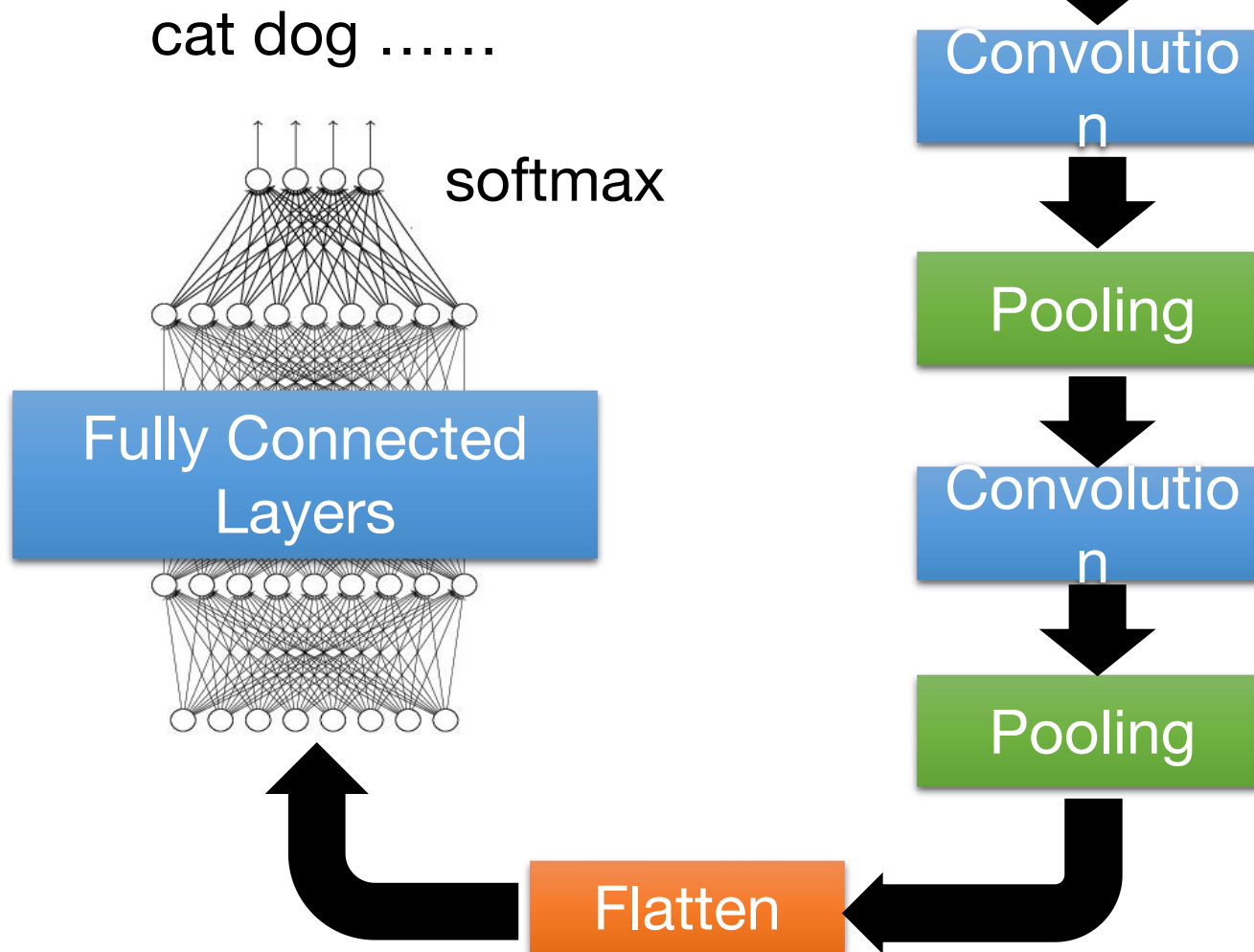
⋮
⋮



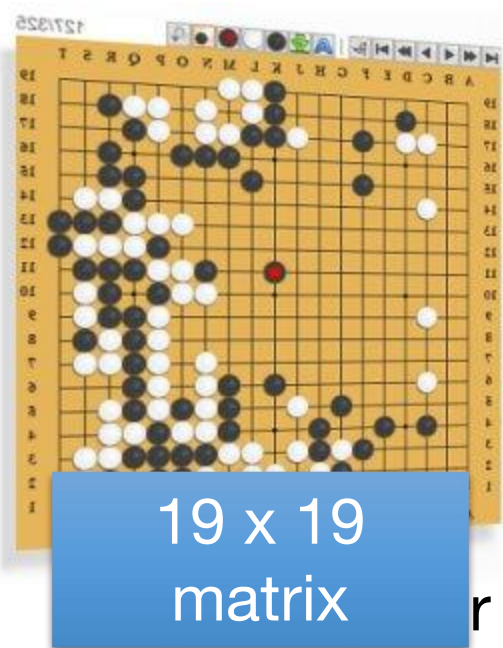
"Image" with 64 channels



The whole CNN



应用: Playing Go



(image)

Black: 1

white: -1

none: 0

48
channels
in Alpha
Go



Next
move
(19 x 19
position
19 x 19 classes)

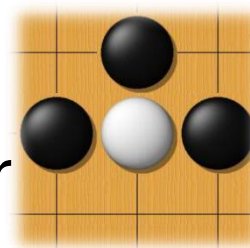
Fully-connected
network can be used

But CNN performs much
better.

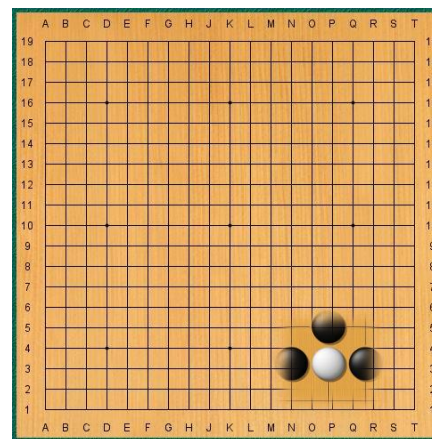
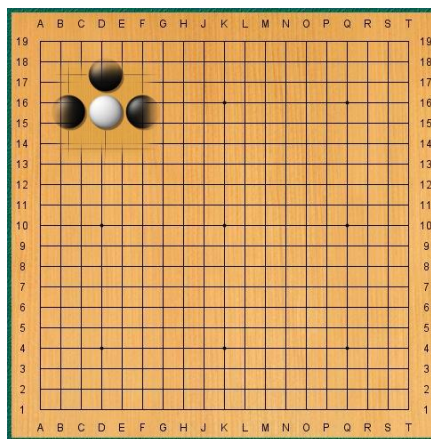
Why CNN for Go playing?

- Some patterns are much smaller than the whole image

Alpha Go uses 5 x 5 for first layer



- The same patterns appear in different regions.



Why CNN for Go playing?

- Subsampling the pixels will not change the



Pooling

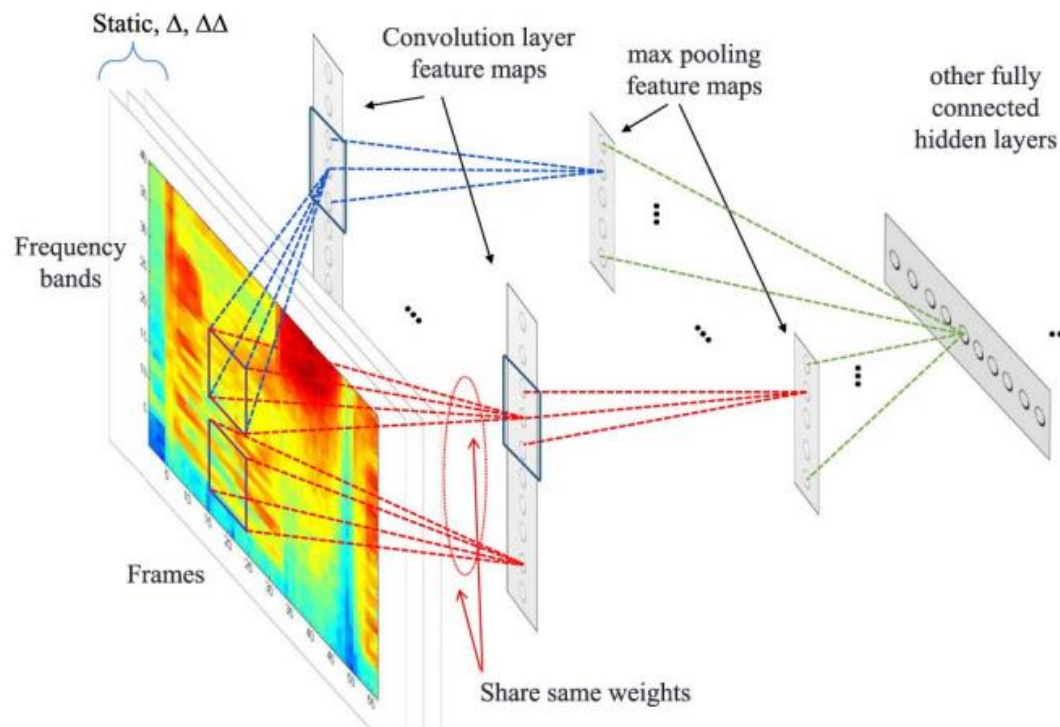
How to explain this???

Neural network architecture. The input to the policy network is a $19 \times 19 \times 48$ image stack consisting of 48 feature planes. The first hidden layer zero pads the input into a 23×23 image, then convolves k filters of kernel size 5×5 with stride 1 with the input image and applies a rectifier nonlinearity. Each of the subsequent hidden layers 2 to 12 zero pads the respective previous hidden layer into a 21×21 image, then convolves k filters of kernel size 3×3 with stride 1, again followed by a rectifier nonlinearity. The final layer convolves 1 filter of kernel size 1×1 with stride 1, with a different bias for each position, and applies a softmax function. The match version of AlphaGo used $k = 192$ filters; Fig. 2b and Extended Data Table 1. The match version of AlphaGo used 256 and 384 filters.

Alpha Go does not use Pooling

Speech

<https://dl.acm.org/doi/10.1109/TASLP.2014.2339736>



Natural Language Processing

<https://www.aclweb.org/anthology/S15-2079/>

